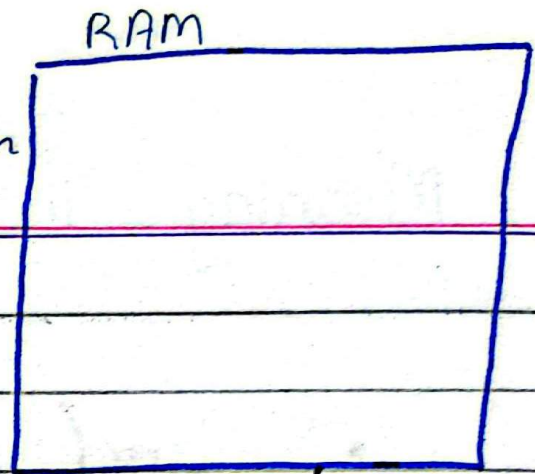


Introduction to computing memory

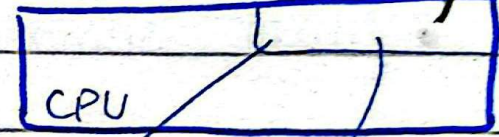
Machine language :- in binary
0 1



Memory, registers, instructions, parameters

very
speedable
but
limited

by which
you modify
data.



register

memory
location
sab sa
teer
memory

jin pa cpu
currently kaam
kar raha hota
us ko
register
ma
rakhna ha

ax bx
cx dx

register
name

ML
- 00100100 -
ADD 0 6

دکٹر میں خاص
وقتیں

it is assembly language

we just translate it in

decimal without any interpreter

1.

2. mov ax, 4

...

jmp 2

Meaning in a language:-

- Syntare (کون سا حرف کسی سے پہلے آئے لہ میں) (کون سے الفاظ آپ بول سکتے ہیں)

(329) - (ab) - (32a)

- Static semantic (

↓
جو بولتی ہیں

if it is syntan
sub ob
I saw a cat - I cat dog. ob
sub verb ob
meaningless else if

- Semantics:- (External word knowledge)

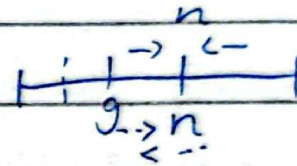
غلطی غلطی غلطی غلطی غلطی
- منسل ہے -
{ I saw a pigeon.
I saw a bravery.

Semantic errors:-

- Crash (every thing dead)

- Run to infinity

e.g sq problem



- Run and produce an output (without assurance of correctness)

↓
very
difficult
error

↓
solved
by
software testing

lets eat grandpa ✗

let's eat, grandpa ✓

Categorization of language:-

low-level

VS

high level

↓
machine

↓
for human

high lev

low lev

(گزارش مرغی use کر لیں)

General

VS

(خاص مقصد کے لیے)

domain specific →

like SQL

use for data base only

slow

one by one
line by line

translate

all at once

high level

interpreted

VS

compiled → run fast

- Source code → (نامی کد لکھتے ہیں) (نام سے بات شروع ہو سکتی)
- Machine code

Python :-

- High level
- General purpose
- Interpreted

C :-

- Medium High level
- General purpose
- Compiled

code لکھنا

- Scripts, programs, shell

REPL

Shell open → command → interpreter →

atom python twitter.com/visualize.html

machine → run → ans

notebook like shell